

Lecture 7

Lattices, LWE, and Regev's Cryptosystem

SEC 595 — Encrypted Computing

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1. Decoding With a Good and a Bad Basis

Everything in this lecture rests on one asymmetry, and it is worth meeting it as a concrete computation before it is dressed up as a cryptosystem.

1.1 Two Bases, One Lattice

Take the two bases

$$B_{\text{bad}} = \left\{ \begin{pmatrix} 6 \\ 14 \end{pmatrix}, \begin{pmatrix} 3 \\ 8 \end{pmatrix} \right\}, \quad B_{\text{good}} = \left\{ \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}.$$

The first is two long, steep, nearly parallel vectors. The second is two short orthogonal ones. The first thing to establish is that they are bases of the *same* lattice, because otherwise the comparison below would be comparing two different problems.

Proposition 1. $\mathcal{L}(B_{\text{bad}}) = \mathcal{L}(B_{\text{good}})$.

Proof. Write each vector of B_{bad} in terms of B_{good} :

$$\begin{pmatrix} 6 \\ 14 \end{pmatrix} = 2 \begin{pmatrix} 3 \\ 0 \end{pmatrix} + 7 \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 8 \end{pmatrix} = 1 \begin{pmatrix} 3 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 2 \end{pmatrix},$$

so $B_{\text{bad}} = B_{\text{good}} U$ with $U = \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$ and $\det U = 8 - 7 = 1$. A matrix of integers with determinant ± 1 is unimodular, hence invertible over $\mathbb{Z}$, so the two bases generate the same set of integer combinations. This is the Lecture 5 criterion, applied. $\square$

Consistently, both bases have the same determinant in absolute value: $6 \cdot 8 - 3 \cdot 14 = 6$ and $3 \cdot 2 - 0 \cdot 0 = 6$. The determinant is an invariant of the lattice — it is the covolume, the area of a fundamental domain — so this was forced.

Concretely, B_{good} makes the lattice legible: it is $\mathcal{L} = 3\mathbb{Z} \times 2\mathbb{Z}$, the set of points with x a multiple of 3 and y a multiple of 2. Nothing in B_{bad} makes that visible, and yet B_{bad} describes the identical set.

1.2 Decoding With the Bad Basis

Give the target

$$v = \begin{pmatrix} 11.6 \\ 4.2 \end{pmatrix},$$

which is not a lattice point, and ask for its expansion in B_{bad} :

$$a \begin{pmatrix} 6 \\ 14 \end{pmatrix} + b \begin{pmatrix} 3 \\ 8 \end{pmatrix} = \begin{pmatrix} 11.6 \\ 4.2 \end{pmatrix} \iff \begin{cases} 6a + 3b = 11.6 \\ 14a + 8b = 4.2. \end{cases}$$

Cramer's rule, with determinant 6, gives

$$a = \frac{11.6 \cdot 8 - 3 \cdot 4.2}{6} = \frac{80.2}{6} = 13.3\bar{6}, \quad b = \frac{6 \cdot 4.2 - 14 \cdot 11.6}{6} = \frac{-137.2}{6} = -22.8\bar{6}.$$

Rounding, $a \mapsto 13$ and $b \mapsto -23$, and substituting back:

$$13 \begin{pmatrix} 6 \\ 14 \end{pmatrix} + (-23) \begin{pmatrix} 3 \\ 8 \end{pmatrix} = \begin{pmatrix} 78 - 69 \\ 182 - 184 \end{pmatrix} = \begin{pmatrix} 9 \\ -2 \end{pmatrix}.$$

The returned point is $(9, -2)$, at Euclidean distance $\sqrt{2.6^2 + 6.2^2} = \sqrt{45.2} \approx 6.72$ from v .

1.3 Decoding With the Good Basis

Now the same target and the same algorithm, with B_{good} :

$$a \begin{pmatrix} 3 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 11.6 \\ 4.2 \end{pmatrix} \implies \begin{array}{llll} 3a = 11.6 & \longrightarrow & a = 3.8\bar{6} & \longrightarrow & a = 4, \\ 2b = 4.2 & \longrightarrow & b = 2.1 & \longrightarrow & b = 2. \end{array}$$

There is no system to solve: the basis is orthogonal, so the matrix is diagonal and each coordinate is one division. Substituting back,

$$4 \begin{pmatrix} 3 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \end{pmatrix},$$

at distance $\sqrt{0.4^2 + 0.2^2} \approx 0.447$ from v . Checking every lattice point in range confirms that $(12, 4)$ is genuinely the closest one: the good basis solved CVP exactly.

1.4 Why the Two Answers Differ

Same lattice, same target, same algorithm — and one basis returns the nearest lattice point while the other returns a point fifteen times further away. The explanation is Proposition 2 plus one number from Lecture 5.

Proposition 2 (When rounding decodes correctly). *Let $t = Bz + r$ with $z \in \mathbb{Z}^n$ and $r \in \mathbb{R}^n$. Then $\text{Babai}_B(t) = Bz$ if and only if*

$$\|B^{-1}r\|_{\infty} < \frac{1}{2}.$$

Definition 1 (Orthogonality defect, Lecture 5). For a basis $B = \{b_1, \dots, b_n\}$ of a lattice of determinant d , the **orthogonality defect** is

$$\delta(B) = \frac{\prod_{i=1}^n \|b_i\|}{|d|} \geq 1,$$

with equality if and only if the b_i are pairwise orthogonal.

Here $\|(6, 14)\| = \sqrt{232} \approx 15.23$ and $\|(3, 8)\| = \sqrt{73} \approx 8.54$, so

$$\delta(B_{\text{bad}}) = \frac{15.23 \times 8.54}{6} \approx 21.69, \quad \delta(B_{\text{good}}) = \frac{3 \times 2}{6} = 1.$$

Now apply Proposition 2. The target is $v = Bz + r$ with the true lattice point $(12, 4)$ and offset $r = (-0.4, 0.2)$, an error of length only 0.447. Measured in each basis:

$$B_{\text{good}}^{-1}r = \begin{pmatrix} -0.4/3 \\ 0.2/2 \end{pmatrix} = \begin{pmatrix} -0.1\bar{3} \\ 0.1 \end{pmatrix}, \quad B_{\text{bad}}^{-1}r = \frac{1}{6} \begin{pmatrix} 8(-0.4) - 3(0.2) \\ -14(-0.4) + 6(0.2) \end{pmatrix} = \begin{pmatrix} -0.6\bar{3} \\ 1.1\bar{3} \end{pmatrix}.$$

In the good basis both coefficients are comfortably inside $(-\frac{1}{2}, \frac{1}{2})$ and rounding recovers z exactly. In the bad basis *both* exceed $\frac{1}{2}$ in absolute value; they round to $(-1, +1)$, so the recovered coefficient vector is $(14, -24) + (-1, 1) = (13, -23)$ — precisely the pair computed in Section 1.3. The failure is not bad luck; it was predictable from the basis alone.

The same statement in Gram–Schmidt terms, which is how Lecture 5 phrased the guarantee: the Gram–Schmidt norms of B_{bad} are 15.23 and 0.394 — their product is the determinant 6, as it must be — so Babai rounding with B_{bad} is only guaranteed for errors shorter than $\frac{1}{2} \cdot 0.394 = 0.197$, and our error has length 0.447. For B_{good} the Gram–Schmidt norms are 3 and 2, giving a guarantee out to radius 1, comfortably beyond 0.447.

	B_{bad}	B_{good}
lattice determinant	6	6
orthogonality defect $\delta(B)$	21.69	1
coefficients of v	(13.37, -22.87)	(3.87, 2.10)
rounded coefficients	(13, -23)	(4, 2)
lattice point returned	(9, -2)	(12, 4)
distance to v	6.72	0.447

The intuition to carry forward: rounding perturbs each coefficient by at most $\frac{1}{2}$, and that half is then *multiplied by a basis vector*. With $(6, 14)$ and $(3, 8)$, half a unit of coefficient error is already several units of position error, and there are two such errors which need not cancel. With $(3, 0)$ and $(0, 2)$ the same half-unit costs at most 1.5 in x and 1 in y , in independent directions.

2. The Naïve Cryptosystem

Section 1 was not a warm-up exercise. It was a decryption attempt, run twice — once with the key an attacker has, once with the key the receiver has. Making that explicit costs three short paragraphs.

2.1 The Key Pair

$$\text{PK} = B_{\text{bad}} = \left\{ \begin{pmatrix} 6 \\ 14 \end{pmatrix}, \begin{pmatrix} 3 \\ 8 \end{pmatrix} \right\}, \quad \text{SK} = B_{\text{good}} = \left\{ \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}.$$

Both bases describe the same lattice, by the proposition of Section 1.1. So the secret is not *which* lattice — that is public — but a convenient description of it. This is the first appearance in the course of a trapdoor that is a change of representation rather than a hidden object.

2.2 Encryption

To encrypt, choose integer coefficients (h, i) , form the lattice point using the **public** basis, and perturb it by a small error:

$$\text{ciphertext} = h b_1 + i b_2 + \vec{e}.$$

With $(h, i) = (14, -24)$:

$$14 \begin{pmatrix} 6 \\ 14 \end{pmatrix} - 24 \begin{pmatrix} 3 \\ 8 \end{pmatrix} = \begin{pmatrix} 84 - 72 \\ 196 - 192 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \end{pmatrix},$$

and with $\vec{e} = (-0.4, 0.2)$ the ciphertext is

$$\begin{pmatrix} 12 \\ 4 \end{pmatrix} + \begin{pmatrix} -0.4 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 11.6 \\ 4.2 \end{pmatrix},$$

which is the target v of Section 1. Note that forming a lattice point needs only *some* basis, and adding a small error needs nothing at all: encryption is available to everyone.

2.3 Decryption

The receiver holds B_{good} and proceeds in two steps.

1. **Decode.** Babai-round the ciphertext in the secret basis. By Section 1.4 this returns the lattice point $(12, 4)$, removing the error.
2. **Read off the message.** Re-express that lattice point in the *public* basis to recover (h, i) . Solving $6h + 3i = 12$, $14h + 8i = 4$ gives

$$h = \frac{12 \cdot 8 - 3 \cdot 4}{6} = 14, \quad i = \frac{6 \cdot 4 - 14 \cdot 12}{6} = -24,$$

exactly what was encrypted. Note that this second solve is exact — there is no rounding — because $(12, 4)$ really is a lattice point.

An attacker holding only B_{bad} can attempt step 1 in that basis instead, and Section 1.3 already computed what happens: the recovered point is $(9, -2)$, whose coefficients in B_{bad} are $(13, -23)$ and not $(14, -24)$. The message is wrong, and it is wrong in both components.

2.4 What the Trapdoor Actually Is

Encryption is easy in any basis; decoding is easy only in a good one.

That asymmetry is the entire construction, and it is worth noticing how little it depends on the specific numbers. Every scheme in the rest of this course is a variation on it.

2.5 Why This Is Only the Naïve Version

Two things are wrong with the scheme as stated, and naming them is what motivates the rest of the lecture.

1. **The arithmetic is real-valued.** Coordinates like 11.6 and errors like -0.4 cannot be transmitted exactly, and any implementation must fix a precision — at which point the security argument has to be redone in whatever discretisation was chosen. We want exact arithmetic from the start, so $\mathbb{Z}_q$ rather than $\mathbb{R}$.

2. **The hardness assumption is about two hand-picked bases.** There is no reason to believe that recovering B_{good} from B_{bad} is hard for *these* matrices, and no way to generate fresh keys with a quantified guarantee. What is needed is a hardness assumption about *random* instances.

Remark. Historically this naïve scheme is not a straw man. It is essentially the GGH cryptosystem of Goldreich, Goldwasser and Halevi (1997), and NTRU is a structured relative of it. Both were broken or badly weakened by attacks that exploited exactly the two defects above — in particular, the error leaked information about the secret basis across many ciphertexts. Regev’s contribution was to replace the ad-hoc trapdoor by an assumption with a worst-case reduction.

3. Learning With Errors

3.1 The Equation

Definition 2 (LWE, Lecture 6). Fix a modulus q , a secret length n , a sample count m , and an error distribution $\mathcal{X}$ over $\mathbb{Z}$ concentrated on small values. An **LWE instance** consists of

- a uniformly random matrix $A \in \mathbb{Z}_q^{m \times n}$,
- a secret vector $s \in \mathbb{Z}_q^n$,
- an error vector $e \in \mathbb{Z}_q^m$ with entries drawn from $\mathcal{X}$,

combined into

$$As + e = b \pmod{q}.$$

Search LWE is the problem of recovering s from (A, b) ; **decision LWE** is the problem of distinguishing $(A, As + e)$ from (A, u) with u uniform. Lecture 6 established that the two are equivalent.

Three objects and one equation. Everything in the remainder of the lecture is built out of that line.

3.2 Delete the Error and It Collapses

Given the equation, the solution is straightforward numerically

$$Ax = b, \quad x = A^{-1}b.$$

It is the whole point, compressed. With $e = 0$ the system $As = b$ is linear over $\mathbb{Z}_q$, and recovering s is Gaussian elimination — polynomial time, completely standard, and taught in first-year linear algebra. Given $m \geq n$ samples it is not even close to hard.

Adding e destroys this. There is no A^{-1} trick that survives an unknown perturbation, because $A^{-1}b = s + A^{-1}e$ and $A^{-1}e$ is not small: A is uniform, so A^{-1} has no reason to have small entries, and multiplying a small vector by it produces something that looks uniform.

Example 1 (Both cases, $q = 7$). Let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & 6 \\ 5 & 2 & 1 \end{bmatrix}, \quad s = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad e = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Then $\det A = 26 \equiv 5 \pmod{7}$, which is invertible, and

$$A^{-1} \equiv \begin{bmatrix} 6 & 1 & 4 \\ 1 & 3 & 0 \\ 3 & 3 & 2 \end{bmatrix} \pmod{7}.$$

Without error, $As \bmod 7 = (6, 1, 5)^T$ and $A^{-1}(6, 1, 5)^T = (1, 2, 3)^T$: the secret comes straight back. With the error, $b = As + e \bmod 7 = (0, 1, 0)^T$ and

$$A^{-1}b \equiv (1, 3, 3)^T \pmod{7},$$

which is not s . One unit of error in one coordinate of b has corrupted the answer, and at $q = 7$ there is no way to tell from (A, b) which of the seven candidates for each coordinate was intended.

4. Regev's Cryptosystem

Throughout, q is the modulus, n the secret length, m the number of samples, and $\bar{b} \in \{0, 1\}$ the single message bit.

4.1 Key Generation

Definition 3 (KeyGen). • **Secret key:** a uniformly random $s \in \mathbb{Z}_q^n$.

• **Public key:** a uniformly random $A \in \mathbb{Z}_q^{m \times n}$ together with

$$b = As + e \bmod q,$$

where $e \in \mathbb{Z}_q^m$ has entries drawn from $\mathcal{X}$.

The key pair is $\text{PK} = (A, b)$ and $\text{SK} = s$.

Three observations. First, A is chosen freely and b is then determined by A , s and e : there is one degree of freedom, not two. Second, the public key is *exactly* m *LWE samples, published together* — so an adversary who could recover s from PK would by definition be solving search LWE. Third, the error e is used once, at key generation, and then discarded. Nobody ever needs it again, including the receiver, who decrypts using s alone.

Remark. Compare the naïve scheme. There the public key was a bad basis and the secret key a good one, and both were bases of the same lattice. Here the public key is m noisy inner products and the secret key is the vector they are inner products *with*. The trapdoor has changed shape — but by Section 3.3, (A, b) still describes a lattice point plus a short error, and s is still what lets you decode it.

4.2 Encryption

Definition 4 ($\text{Enc}(\text{PK}, \bar{b})$). To encrypt a bit $\bar{b}$:

1. choose a uniformly random binary vector $x \in \{0, 1\}^m$;
2. compute

$$c_1 = A^T x \bmod q \in \mathbb{Z}_q^n, \quad c_2 = b^T x + \bar{b} \lfloor q/2 \rfloor \bmod q \in \mathbb{Z}_q.$$

The ciphertext is the pair (c_1, c_2) .

What is happening: the random x selects a subset of the published samples and adds them up. A subset-sum of LWE samples is itself an LWE sample — with secret s , coefficient vector $A^T x$ and error $e^T x$ — so the encryptor has manufactured a *fresh* sample nobody has seen, without knowing s . The message is then added to the scalar half, shifted to the far side of the modulus by the factor $\lfloor q/2 \rfloor$.

Note the division of labour: c_1 carries no message at all. It exists so that the receiver can reconstruct, and subtract, the masking term hidden inside c_2 .

Forward reference. This is also where the security proof lives, and it is worth flagging even though we do not give it here. Decision LWE says the public key (A, b) is indistinguishable from uniform; a leftover-hash argument then says that the pair $(A^T x, b^T x)$ is close to uniform as well, for random binary x . So the ciphertext hides $\bar{b}$ by being statistically close to a uniform pair, and the scheme is IND-CPA secure. Lecture 9 defines IND-CPA properly, alongside the other security notions, and explains why a *randomised* encryption — which Regev is, through x — is a precondition for any of them. The advantage and negligibility vocabulary of Lecture 1, Section 8 is what the statement is written in.

4.3 Decryption

Definition 5 ($\text{Dec}(s, (c_1, c_2))$). The receiver, who knows s , computes

$$\Delta = c_2 - c_1^T s \bmod q,$$

and returns

$$\bar{b} = \begin{cases} 1 & \text{if } \Delta \text{ is closer to } \lfloor q/2 \rfloor, \\ 0 & \text{if } \Delta \text{ is closer to } 0, \end{cases}$$

where “closer” means smaller distance *around the circle* $\mathbb{Z}_q$, i.e. $\text{dist}(\alpha, \gamma) = \min\{|\alpha - \gamma|, q - |\alpha - \gamma|\}$.

The entire cost of decryption is one inner product of length n and one subtraction.

Two features of the rule deserve emphasis. It does not test Δ for equality with anything — it asks which of two landmarks, 0 and $q/2$, the value is nearer to. And the distance is measured around the circle, because $\mathbb{Z}_q$ wraps: in the instance of Section 6, $\Delta = 6$ with $q = 7$ is at distance 1 from 0, not 6.

5. Correctness and the Rounding Decision

Section 4 defined the scheme. It did not say why decryption returns the bit that was encrypted, and that is the only thing left to prove.

5.1 What Δ Actually Equals

Proposition 3. *For any key pair and any ciphertext produced by Enc,*

$$\Delta = \bar{b} \lfloor q/2 \rfloor + e^T x \pmod{q}.$$

Proof. Substitute the definitions. With $c_1 = A^T x$ and $b = As + e$, all modulo q :

$$\begin{aligned} c_1^T s &= (A^T x)^T s = x^T As, \\ b^T x &= (As + e)^T x = s^T A^T x + e^T x = x^T As + e^T x, \end{aligned}$$

the last step because $s^T A^T x$ is a scalar and hence equal to its own transpose $x^T A s$. Subtracting,

$$\Delta = c_2 - c_1^T s = (x^T A s + e^T x + \bar{b} \lfloor q/2 \rfloor) - x^T A s = \bar{b} \lfloor q/2 \rfloor + e^T x. \quad \square$$

The message survives untouched. What is left over is the error, weighted by the encryptor's random bits — a single scalar, which we will call $\eta = e^T x$.

5.2 Why the Cancellation Happens at All

It is worth pausing on the structure of that proof, because the same shape recurs throughout the course.

Neither party can compute $x^T A s$ on its own: the encryptor knows x and A but not s ; the receiver knows s and A but not x . Each nonetheless computes *one side* of that quantity — the encryptor inside $b^T x$, the receiver inside $c_1^T s$ — and the subtraction destroys it. The masking term is never transmitted and never known in full by either party.

This is the same shape as Diffie–Hellman, where both parties arrive at g^{ab} without either sending it. The difference is that here the agreement is only approximate: the two sides differ by $e^T x$, and the rest of this section is about making that discrepancy harmless.

5.3 Why the Rounding Decision Works

By Proposition 3,

$$\Delta = \begin{cases} \eta & \text{if } \bar{b} = 0, \\ \lfloor q/2 \rfloor + \eta & \text{if } \bar{b} = 1, \end{cases} \quad \eta = e^T x.$$

So Δ sits near 0 in one case and near $q/2$ in the other, displaced by the same η in both. The two landmarks are half the modulus apart, so each has a basin of radius $q/4$; asking which is nearer returns $\bar{b}$ provided η never pushes Δ out of its own basin.

Centred value of Δ	Nearer landmark	Decoded bit
$ \lfloor \Delta \rfloor_q < q/4$	0	0
$ \lfloor \Delta \rfloor_q > q/4$	$q/2$	1

5.4 The Correctness Condition

Theorem 1 (Correctness of Regev encryption). *Let $\eta = [e^T x]_q$ be the centred representative. If q is even and*

$$|\eta| < \frac{q}{4},$$

then $\text{Dec}(s, \text{Enc}(PK, \bar{b})) = \bar{b}$ for both values of $\bar{b}$. If q is odd the same conclusion holds under $|\eta| + \frac{1}{2} < q/4$.

Proof. Write $\lfloor q/2 \rfloor = q/2 - \varepsilon$ with $\varepsilon = 0$ for even q and $\varepsilon = \frac{1}{2}$ for odd q .

If $\bar{b} = 0$ then $\Delta \equiv \eta$, so $\text{dist}(\Delta, 0) = |\eta|$ and $\text{dist}(\Delta, q/2) = q/2 - |\eta|$. The former is smaller exactly when $|\eta| < q/4$, and the rule returns 0.

If $\bar{b} = 1$ then $\Delta \equiv q/2 + (\eta - \varepsilon)$. Setting $\zeta = \eta - \varepsilon$ and assuming $|\zeta| \leq q/2$,

$$\text{dist}(\Delta, q/2) = |\zeta|, \quad \text{dist}(\Delta, 0) = \frac{q}{2} - |\zeta|,$$

so the rule returns 1 exactly when $|\zeta| < q/4$. Since $|\zeta| \leq |\eta| + \varepsilon$, both cases are covered by the stated hypotheses. $\square$

Remark. We will quote this condition throughout the course in the simpler form $|e^T x| < q/4$, dropping the odd- q half-unit. That is standard, and harmless: in any real parameter set q is at least 2^{12} , so a half-unit of slack is 2^{-11} of the budget. It is worth remembering that the half is there, because in the $q = 7$ instance of Section 6 the whole budget is 1.75.

Next, turn the condition into something one can check at parameter-selection time. Lecture 6's bound $\|e\|_\infty \leq \beta$ does this in one step.

Corollary 1 (The parameter inequality). *Since $x \in \{0, 1\}^m$ has at most m nonzero entries, $|e^T x| \leq m\beta$, and decryption is correct whenever*

$$m\beta < \frac{q}{4}.$$

Two things keep $e^T x$ small, and both are design choices rather than accidents. The entries of e are small by construction, drawn from $\mathcal{X}$; and x is *binary*, so $e^T x$ is a sum of *some* of the entries of e and never a scaled-up multiple of them. Had x been allowed uniform entries in $\mathbb{Z}_q$, the bound would have been $mq\beta$ and nothing would decrypt. Two things make it large: a big m , and an error distribution with too much spread.

Remark (Worst case versus typical case). Corollary 1 is a worst-case bound and is very pessimistic. For e with independent entries of standard deviation σ and uniform binary x , the sum $e^T x$ has standard deviation about $\sigma\sqrt{m/2}$ — growing like $\sqrt{m}$, not m . Real parameter sets choose q so that a 6σ -style tail bound clears $q/4$, accepting a negligible decryption failure probability rather than paying for the worst case. This trade — a $\sqrt{m}$ heuristic bound against an m certain one — is the single most consequential choice in lattice parameter selection.

Remark (The Lecture 1 layout, recovered). Set $t = 2$ in the Lecture 1 picture: plaintext modulus t , ciphertext modulus q , scaling factor $\lfloor q/t \rfloor$. Then $\lfloor q/2 \rfloor$ is exactly that scaling factor, and $\Delta = \bar{b}\lfloor q/2 \rfloor + \eta$ is exactly the layout “message in the high bits, noise in the low bits”. Regev is the $t = 2$ case of the encoding that BFV will use for general t , and the decision rule of Section 4.3 is the $t = 2$ case of $\lfloor t \cdot \Delta/q \rfloor$.

Forward reference. Regev encrypts one bit, and says nothing about computing on ciphertexts. Lecture 8 writes down what a homomorphic scheme would have to provide — a fourth algorithm, two correctness statements and one commuting diagram — and Lecture 9 exhibits two schemes (ElGamal and Paillier) that each provide half of it. The LWE machinery of this lecture returns in Lecture 12 as **GSW**, where the ciphertext becomes a matrix and the secret becomes an approximate eigenvector; there the leftover term $e^T x$ stops being a quantity bounded once and becomes one that must be tracked through every operation. Lecture 11's bootstrapping is the answer to what happens when that tracking runs out of budget.

Forward reference. Regev is also the direct ancestor of **ML-KEM**, the NIST key-encapsulation standard of FIPS 203. Three changes separate them. The vectors and matrices over $\mathbb{Z}_q$ become vectors and matrices over the ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ of Lecture 3, which is what makes A compact and multiplication an NTT rather than a matrix product; the single-bit payload becomes a 256-bit key, encoded one bit per ring coefficient by exactly the $\lfloor q/2 \rfloor$ trick of Section 4.2; and the low-order bits of the ciphertext are dropped, which adds a rounding error to the same noise

budget analysed above. The decapsulation of ML-KEM is, at its core, $\Delta = c_2 - c_1^T s$ followed by a rounding decision — with $q = 3329$, the modulus from Lecture 1.

6. A Worked Example

One complete instance, small enough to check by hand, with every value reduced modulo q at each step. The parameters are

$$q = 17, \quad n = 2, \quad m = 4, \quad \mathcal{X} \text{ uniform on } \{-1, 0, 1\} \ (\beta = 1), \quad \lfloor q/2 \rfloor = 8.$$

They are chosen so that Corollary 1 holds: $m\beta = 4 < q/4 = 4.25$. Decryption is therefore guaranteed for *every* choice of e and x , not merely for the ones below.

6.1 Key Generation: Four LWE Samples

The receiver picks the secret and a uniform matrix:

$$s = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad A = \begin{bmatrix} 4 & 11 \\ 7 & 2 \\ 13 & 6 \\ 1 & 15 \end{bmatrix}.$$

Row by row,

$$As = \begin{bmatrix} 4 \cdot 3 + 11 \cdot 5 \\ 7 \cdot 3 + 2 \cdot 5 \\ 13 \cdot 3 + 6 \cdot 5 \\ 1 \cdot 3 + 15 \cdot 5 \end{bmatrix} = \begin{bmatrix} 67 \\ 31 \\ 69 \\ 78 \end{bmatrix} \equiv \begin{bmatrix} 16 \\ 14 \\ 1 \\ 10 \end{bmatrix} \pmod{17}.$$

With the error $e = (1, -1, 0, 1)^T$ drawn from $\mathcal{X}$,

$$b = As + e \pmod{17} = \begin{bmatrix} 16 + 1 \\ 14 - 1 \\ 1 + 0 \\ 10 + 1 \end{bmatrix} \pmod{17} = \begin{bmatrix} 0 \\ 13 \\ 1 \\ 11 \end{bmatrix}.$$

So $\text{PK} = (A, b)$ with $b = (0, 13, 1, 11)^T$, and $\text{SK} = s = (3, 5)^T$. The public key is four LWE samples (a_i, b_i) , each satisfying $b_i = \langle a_i, s \rangle + e_i \pmod{17}$ with $e_i \in \{-1, 0, 1\}$; the error is now discarded. An attacker holding (A, b) faces exactly the problem of Section 3.2: with e deleted, two of the equations would give s by elimination, but with e present every candidate must be tested against all four residuals.

6.2 Encrypting $\bar{b} = 1$

The sender draws the binary vector $x = (1, 0, 1, 1)^T$, which selects samples 1, 3 and 4. Adding those rows of A , and those entries of b ,

$$c_1 = A^T x = \begin{bmatrix} 4 + 13 + 1 \\ 11 + 6 + 15 \end{bmatrix} = \begin{bmatrix} 18 \\ 32 \end{bmatrix} \equiv \begin{bmatrix} 1 \\ 15 \end{bmatrix} \pmod{17}, \quad b^T x = 0 + 1 + 11 = 12,$$

$$c_2 = b^T x + \bar{b} \lfloor q/2 \rfloor = 12 + 8 = 20 \equiv 3 \pmod{17}.$$

The ciphertext is $(c_1, c_2) = ((1, 15)^T, 3)$. The sender never used s : $(c_1, c_2 - 8)$ is a fresh LWE sample with coefficient vector c_1 and error $e^T x = e_1 + e_3 + e_4 = 2$.

6.3 Decrypting

The receiver computes

$$c_1^T s = 1 \cdot 3 + 15 \cdot 5 = 78 \equiv 10 \pmod{17}, \quad \Delta = c_2 - c_1^T s = 3 - 10 = -7 \equiv 10 \pmod{17}.$$

Around the circle $\mathbb{Z}_{17}$, 10 is at distance 2 from $\lfloor q/2 \rfloor = 8$ and at distance $\min\{10, 7\} = 7$ from 0, so the rule of Section 4.3 returns $\bar{b} = 1 \checkmark$. Equivalently, the centred representative is $[10]_{17} = -7$ and $|-7| > 17/4$.

Proposition 3 predicted this value exactly:

$$\Delta = \bar{b} \lfloor q/2 \rfloor + e^T x = 8 + 2 = 10.$$

The masking term $x^T A s$ was computed by both parties — by the sender inside $b^T x$, by the receiver inside $c_1^T s$ — and cancelled in the subtraction. What survives is the message, displaced by the noise $\eta = 2$, well inside its basin of radius $q/4 = 4.25$.

6.4 Encrypting $\bar{b} = 0$, and the Wraparound

Encryption is randomised, so a second message uses fresh bits. Take $x' = (0, 1, 1, 0)^T$, selecting samples 2 and 3:

$$c'_1 = \begin{bmatrix} 7 + 13 \\ 2 + 6 \end{bmatrix} \equiv \begin{bmatrix} 3 \\ 8 \end{bmatrix} \pmod{17}, \quad c'_2 = b^T x' + 0 = 13 + 1 = 14.$$

Decrypting,

$$c'^T_1 s = 3 \cdot 3 + 8 \cdot 5 = 49 \equiv 15 \pmod{17}, \quad \Delta' = 14 - 15 = -1 \equiv 16 \pmod{17}.$$

Read naively, 16 looks far from 0. Around the circle it is at distance 1 from 0 and at distance 8 from 8, so the rule returns $\bar{b} = 0 \checkmark$. Again Proposition 3 agrees: $\Delta' = 0 + e^T x' = e_2 + e_3 = -1 + 0 = -1$. This is why distance must be measured around $\mathbb{Z}_q$: a negative noise lands just below q , not just above 0.

Remark (What the instance shows). Both decryptions left a margin: $|\eta| = 2$ and $|\eta'| = 1$ against a budget of 4.25. Over all binary x the largest value for this particular e is $|e^T x| = 2$ (select e_1 and e_4), and for any e from $\mathcal{X}$ it is at most $m\beta = 4$; both are below $q/4$ — which is Theorem 1 and Corollary 1 at work. Doubling the error range to $\{-2, \dots, 2\}$ would give $m\beta = 8 > 4.25$, and some choices of e and x would then decrypt incorrectly.

Summary

1. A lattice has infinitely many bases, all with the same determinant, and related by unimodular change of basis. Babai rounding decodes a perturbed lattice point by writing the target in the basis and rounding; it is correct exactly when the offset, *measured in that basis*, has all coefficients below $\frac{1}{2}$ in absolute value.
2. That criterion is a statement about the basis, not the error. A basis with orthogonality defect 21.7 fails on an error of length 0.45 that a basis with defect 1 handles easily. Rounding error is multiplied by the basis vectors.

3. The asymmetry is a trapdoor: publish a bad basis, keep a good one. Anyone can form a noisy lattice point; only the key holder can decode it. The naïve scheme built on this is essentially GGH, and it fails for two reasons — real arithmetic, and no hardness assumption about random instances.
4. Learning With Errors is the same statement in $\mathbb{Z}_q$. In $As + e = b$ the term As is a point of the q -ary lattice $\Lambda_q(A)$ and e is short, so recovering s is BDD. Delete e and the problem collapses to Gaussian elimination.
5. **Regev.** SK = s ; PK = (A, b) with $b = As + e$. Encrypt a bit as $c_1 = A^T x$, $c_2 = b^T x + \bar{b} \lfloor q/2 \rfloor$ for uniform binary x . Decrypt as $\Delta = c_2 - c_1^T s$ and round. The public key is m LWE samples; the ciphertext is one fresh sample the encryptor manufactured without knowing s .
6. Correctness is a single cancellation: $\Delta = \bar{b} \lfloor q/2 \rfloor + e^T x$, because both parties compute the masking term $x^T As$ and the subtraction destroys it. Neither party can compute that term alone — the same shape as Diffie–Hellman.
7. The bit is recovered by asking whether Δ is nearer 0 or $q/2$ around the circle. That is correct whenever $|e^T x| < q/4$, hence whenever $4m\beta < q$: choose q large relative to the noise. The worst-case bound grows like m , the typical value like $\sqrt{m}$, and parameter selection lives in the gap.
8. The message occupies the top of the modulus and the noise the bottom. That is the $t = 2$ case of the $\Delta m + e$ layout introduced in Lecture 1, and it is the layout every scheme in the rest of the course will reuse.

Exercises

1. Solve $6a + 3b = 11.6$, $14a + 8b = 4.2$ exactly, as fractions. Round each coefficient and compute the resulting lattice point. Confirm that the scan's second-colour reading $a = 14$ is not what the arithmetic gives, and compute the point that reading *would* have produced.
2. Verify directly that B_{bad} and B_{good} generate the same lattice by exhibiting the integer matrix U with $B_{\text{bad}} = B_{\text{good}} U$ and computing $\det U$; then write $(9, -2)$ and $(15, 12)$ in the basis B_{good} to confirm both are lattice points. Compute the orthogonality defect $\delta(B)$ of each basis. Finally, apply Proposition 2 to the offset $r = (-0.4, 0.2)^T$: compute $B^{-1}r$ in each basis and confirm that the criterion predicts success for one and failure for the other *before* any rounding is done.
3. Find an offset r of Euclidean length less than 0.25 for which Babai rounding in B_{bad} nevertheless fails, and determine the exact infimum of $\|r\|$ over all failing offsets. (Hint: the second row of B_{bad}^{-1} is $(-\frac{7}{3}, 1)$; make r parallel to it. Compare your infimum with half the smallest Gram–Schmidt norm, $0.394/2$.) What does this show about bounding decoding error by $\|r\|$ alone?
4. Run the naïve cryptosystem end to end: encrypt $(h, i) = (14, -24)$ with error $\vec{e} = (-0.4, 0.2)^T$, decrypt with B_{good} and recover (h, i) ; then repeat the decryption with B_{bad} in place of B_{good} and say precisely which of the two steps of Section 2.3 goes wrong, and what wrong message results.
5. With A and s as in Section 3.2 and $q = 7$: verify that $\det A \equiv 5$, compute $A^{-1} \bmod 7$, and confirm that $A^{-1}(As \bmod 7) = s$. Now compute $A^{-1}b$ for the noisy $b = (0, 1, 0)^T$ and report

how many coordinates of s it recovers. How many secrets $s' \in \mathbb{Z}_7^3$ are consistent with this (A, b) if the only promise is $\|e\|_\infty \leq 2$?

6. For the instance of Section 6, compute c_2 and the ciphertext Alice should send, then compute $c_1^T s$ both before and after reduction modulo 7 and confirm that Δ is the same either way. Verify your Δ against Proposition 3. Which of the three conflicting values in the scan — summand 4, total 3, transmitted 4 — is the correct one?
7. Apply the decision rule of Section 4.3 to your Δ from Exercise 7, measuring distance around the circle $\mathbb{Z}_7$. Which bit does it return, and which was encrypted? Then repeat the whole instance with $\bar{b} = 0$ and show that decryption fails in that case too. Reconcile both failures with Theorem 1.
8. Keeping s , e , x and A exactly as they are, find the smallest modulus q for which the instance decrypts correctly for *both* values of $\bar{b}$.
 - (a) Predict the answer from the condition $|e^T x| < q/4$ before computing anything.
 - (b) Confirm it by evaluating Δ for both bits at each candidate q from 7 upwards.
 - (c) Show that $q = 12$ produces an exact tie for $\bar{b} = 0$, and explain what this says about the strictness of the inequality in Theorem 1.

Then state in one sentence why $q = 7$ was never going to be enough.

9. Take $n = 4$, $m = 8$, and errors drawn uniformly from $\{-1, 0, 1\}$.
 - (a) What is the worst-case value of $|e^T x|$ over all binary x , and what is the smallest q that Corollary 1 then guarantees?
 - (b) Compute the standard deviation of $e^T x$ for uniform e and uniform binary x , and say what q a six-sigma criterion would suggest instead. You should find that at $m = 8$ it suggests a *larger* q than part (a) did. Explain why that is not a contradiction.
 - (c) The worst-case bound grows like m and the six-sigma bound like $\sqrt{m}$. Find the smallest m at which the six-sigma criterion becomes the cheaper of the two, and say what this implies about which bound real parameter sets use.
 - (d) Now suppose the encryptor chose x uniformly from $\mathbb{Z}_q^m$ instead of from $\{0, 1\}^m$. Show that Proposition 3 is unaffected, that the bound of Corollary 1 degrades to $|e^T x| \leq mq\beta/2$, and hence that it never clears $q/4$ for any q , whatever β . Argue informally that $e^T x$ really is close to uniform modulo q , so the failure is not an artefact of a loose bound. Which property of the binary x was doing the work?
10. Prove the “odd q ” half of Theorem 1 carefully: show that using $\lceil q/2 \rceil$ in place of $\lfloor q/2 \rfloor$ changes the sufficient condition for $\bar{b} = 1$ to $|\eta + \frac{1}{2}| < q/4$, and exhibit an η and an odd q for which one convention decrypts correctly and the other does not. (One such pair uses the modulus of Section 6.)